

ProgressThesisJuly Part 2

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1 Warnings

This document is written in a colloquial way.

2 Introduction

As I told you in the previous document `ProgressThesisJuly.pdf`, I had a few **non-study hyperparameters** left to be analyzed, in particular: **threshold density**, **warm up time**, **time interval heuristic**. Following its analysis are presented.

3 Completed work

3.1 `threshold_density`

To explain this, I will include the docstring of the script that performs the sensitivity analysis for this hyperparameter.

```
"""
One-off tool: sweeps a range of threshold_density values and plots the
final R-gap for each, to determine whether this imputation hyperparameter
meaningfully affects the obtained R-gap and, if so, which value minimises it.

threshold_density governs how missing cells in the average link travel time
table are imputed before TDSP computation. The table spans
(episode edge time_interval); cells are missing wherever no vehicle
traversed an edge in a given interval and episode. Two strategies exist:

    density > threshold_density  forward-fill from the previous interval
                                (signal: congestion was present, recent
                                observation is a better proxy than free-flow)
    density <= threshold_density free-flow travel time
                                (signal: edge was essentially empty, so
                                free-flow is the appropriate cost)

A threshold of 0 means forward-fill is applied whenever any vehicle was
recorded on the edge in that interval; higher thresholds require stronger
congestion evidence before forward-fill is preferred over free-flow.

The metric plotted:

    Final R-gap  R-gap at the last training episode. Lower is better.
    A higher value means drivers collectively lose more time by not switching
    to their time-dependent shortest path.

Decision rule: Choose the threshold_density that yields the lowest final R-gap.
If the curve is flat, the hyperparameter has negligible impact and any value
(e.g. the default of 0) can be kept.
```

```
Final decision: 10
(R-gap varies substantially for threshold values below approximately 5 vehicles/km/lane.
Beyond 10 vehicles/km/lane, further increases had a negligible effect on the R-gap.)
```

```
Run with: python src/tools/sensitivity/plot_threshold_density.py <config.yaml>
```

```
Parameter dependencies: Independent hyperparameter
(The threshold_density parameter is considered an independent hyperparameter.
Although its impact on the computed R-gap depends on factors such as the temporal
aggregation interval, network congestion level, and the proportion of missing
observations, these factors only influence the frequency with which the
imputation rule is applied. They do not alter the decision rule itself)
"""
```

I can see in both demand scenarios, that the R-gap varies substantially for threshold values below approximately 5 vehicles/km/lane. Beyond 10 vehicles/km/lane, further increases had a negligible effect on the R-gap.

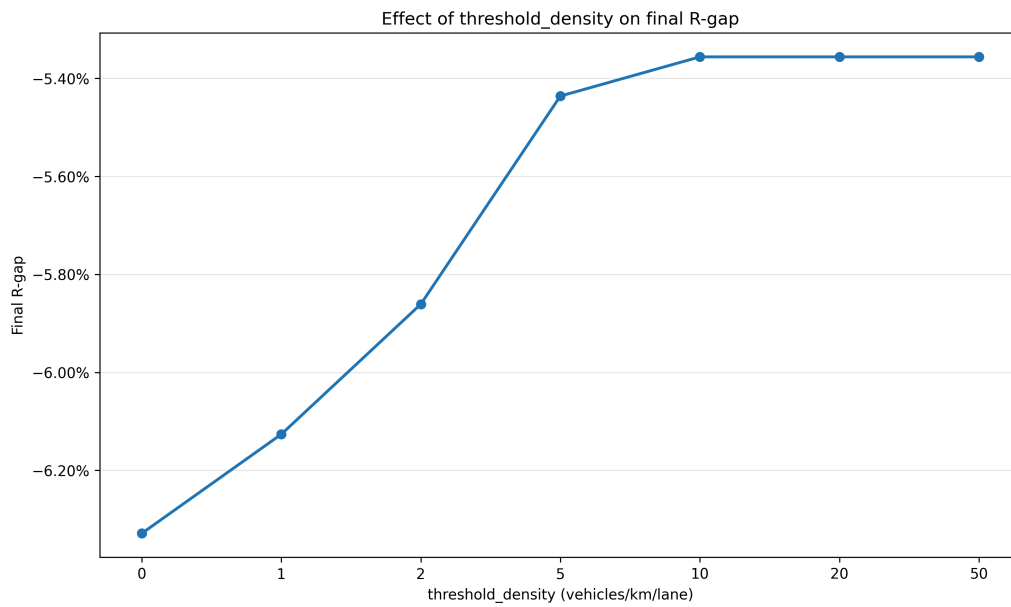


Figure 1: Sensitivity analysis threshold_density hyperparameter. **Demand: 1000 vehicles**

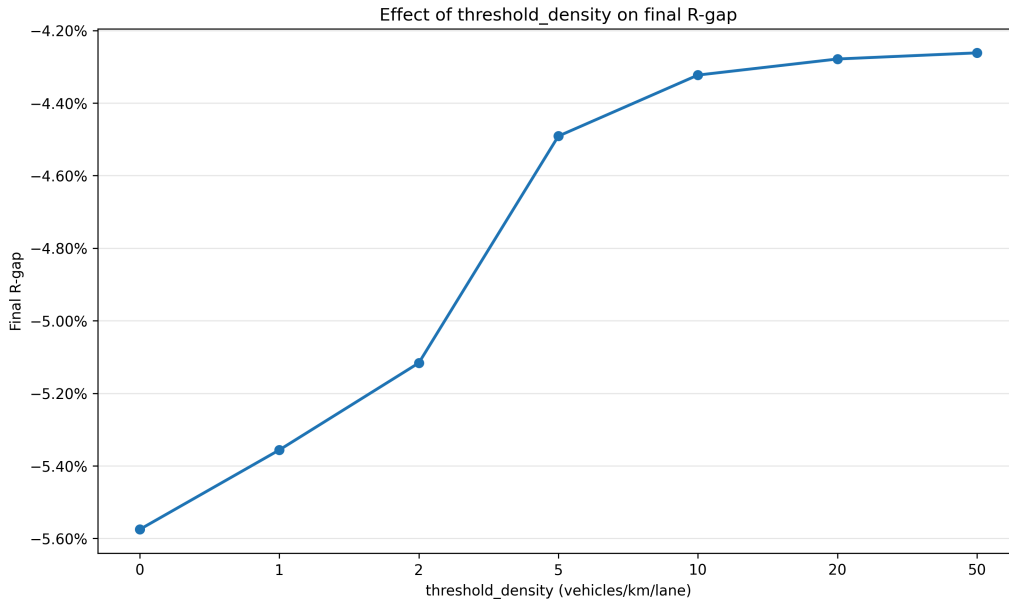


Figure 2: Sensitivity analysis threshold_density hyperparameter. **Demand: 1500 vehicles**

Final decision 10 vehicles

3.2 warm_up_time

To explain this, I will include the docstring of the script that performs the sensitivity analysis for this hyperparameter.

```
"""
One-off tool to determine the minimum warm-up time required for the
initial transient caused by starting the simulation from an empty network
to disappear.

Context
-----
The simulation always starts with an empty network. Consequently, the
first vehicles experience unrealistically good traffic conditions (no
queues and little or no congestion), making them unrepresentative of the
steady operating conditions of interest.

A warm-up period is therefore introduced so that only vehicles entering
after the network has reached a representative traffic state are
evaluated.

Methodology
-----
For each representative demand level:

1. Calibrate the demand.
2. Run only the first SUMO episode using a random routing policy.
3. Compute the number of vehicles in the network every second and plot the resulting curve
4. Select the smallest warm-up candidate after which the initial
   increasing trend associated with the network filling process has
   disappeared.

Only the first episode is analyzed because it isolates the transient
caused by starting from an empty network. In later episodes, changes in
the occupancy curve may also reflect improvements in the learned routing
policy, making the initialization transient more difficult to interpret.

Metric
```

```

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The number of vehicles in the network is used because it directly
captures the phenomenon that motivates the warm-up. Since the network
starts empty, the occupancy curve provides the most direct indication of
when the initial filling transient has ended.

```

```

Final decision
-----

```

```

A warm-up time of 5 minutes was selected because it is the smallest
candidate after which all considered demand scenarios exhibit a
stable occupancy level rather than the initial increasing trend.

```

```

Dependencies
-----

```

```

The required warm-up time depends on:

```

- traffic demand,
- network topology.
- vehicle departure schedule (uniform in all experiments).

```

Run
---
```

```

python src/tools/sensitivity/plot_warm_up.py <config.yaml>
"""

```

A warm-up time of 5 minutes was selected because it is the smallest candidate after which all considered demand scenarios exhibit a stable occupancy level rather than the initial increasing trend.

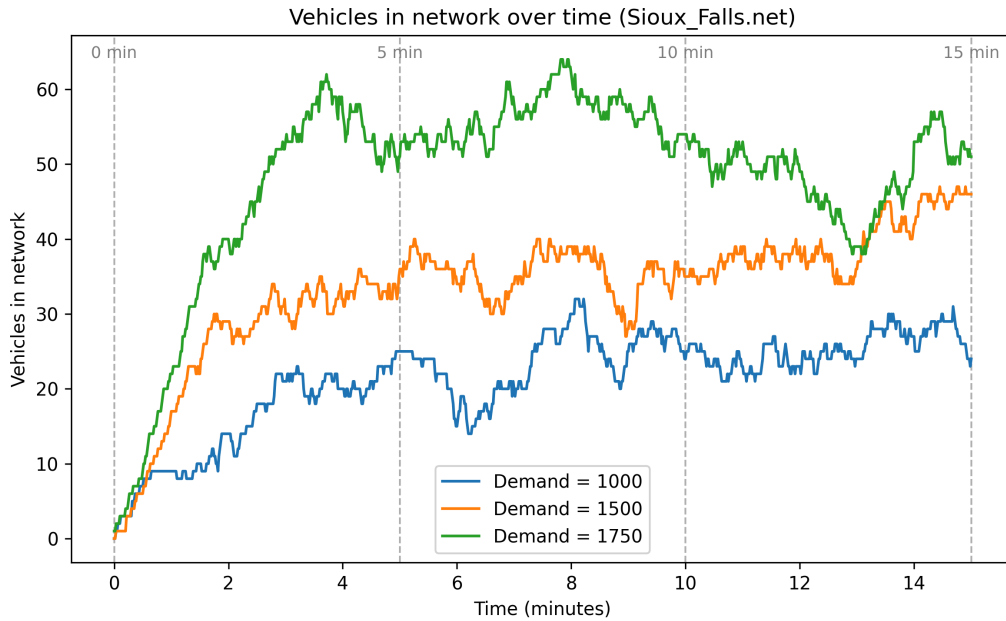


Figure 3: Sensitivity analysis warm_up_time hyperparameter.

Final decision 5 min

3.3 fixed_time_min

To explain this, I will include the docstring of the script that performs the sensitivity analysis for this hyperparameter.

```

"""
One-off tool: sweeps a range of fixed_time_min values and plots, for each
demand scenario, the full R-gap convergence curve and the aggregate
missingness of the average link travel time table, to determine whether

```

this aggregation hyperparameter meaningfully affects the R-gap and, if so, which value to choose.

`fixed_time_min` is an hyperparameter that affects the computation of the R-gap. It does not alter the MARL algorithm, only the R-gap evaluation. It governs the width of the time intervals used to aggregate edge travel times and to assign departure-interval labels to agents. It enters the R-gap computation through two competing effects:

Smaller interval	higher temporal resolution and more accurate average travel-time estimates, but fewer observations per cell, leading to more missing values and therefore more imputation.
Larger interval	denser observations per cell (fewer missing values and less imputation), but greater temporal aggregation resulting in less accurate travel-time estimates.

Metrics plotted (one figure per demand scenario):

R-gap convergence	full R-gap curve across episodes, one line per <code>fixed_time_min</code> candidate.
Missingness (%)	proportion of missing cells in the average link travel time table (episode edge time_interval), one point per candidate.

Within the range of intervals considered, improving the temporal accuracy of the travel-time estimates has a larger positive effect on the R-gap than the negative effect introduced by the additional imputation. However, when the interval becomes excessively small, the amount of missing data starts to dominate.

The candidate intervals were explored progressively, starting from coarse intervals (minutes) and refining the search towards shorter intervals (seconds) once the general trend had been identified.

Decision rule:

Select the smallest interval whose R-gap curve remains as close to zero as possible while avoiding frequent negative R-gap values.

Final decision:

15 seconds (0.25 minutes).

Run with: `python src/tools/sensitivity/plot_fixed_time_min.py <config.yaml>`

Parameter dependencies: `fixed_time_min` interacts with network size (more edges more potential missing cells) and traffic demand (higher demand denser observations, and potentially less missing values because we have more vehicles in the network).

"""

The `fixed_time_min` hyperparameter determines the length of the time intervals used throughout the computation of the R-gap. It affects the R-gap through two competing effects:

- Smaller time intervals provide more accurate average travel-time estimates. However, they also increase the number of missing values that must be imputed in the average link travel time table.
- Larger time intervals reduce the amount of missing data. However, the increased temporal aggregation produces less accurate average travel-time estimates.

The sensitivity analysis suggests that, within the considered range, the benefit of improving the accuracy of the average travel-time estimates outweighs the negative effect of the additional missing values (hence smaller intervals will be preferred). Nevertheless, if the interval becomes excessively small, the large amount of missing data begins to dominate, degrading the quality of the R-gap computation.

Increasing the time interval progressively shifts the R-gap towards negative values, whereas decreasing the time interval shifts it towards positive values.

Finally, as expected, higher demand levels produce fewer missing values because more vehicles traverse the network

3.3.1 First group of plots

The first experiment considers time intervals of 5, 10, 15, 20 and 30 minutes.

Among these candidates, the 5-minute interval consistently produces R-gap values closest to zero for most demand scenarios. Furthermore, the results reveal a clear trend: increasing the time interval shifts the R-gap towards increasingly negative values.

The corresponding results are shown below.

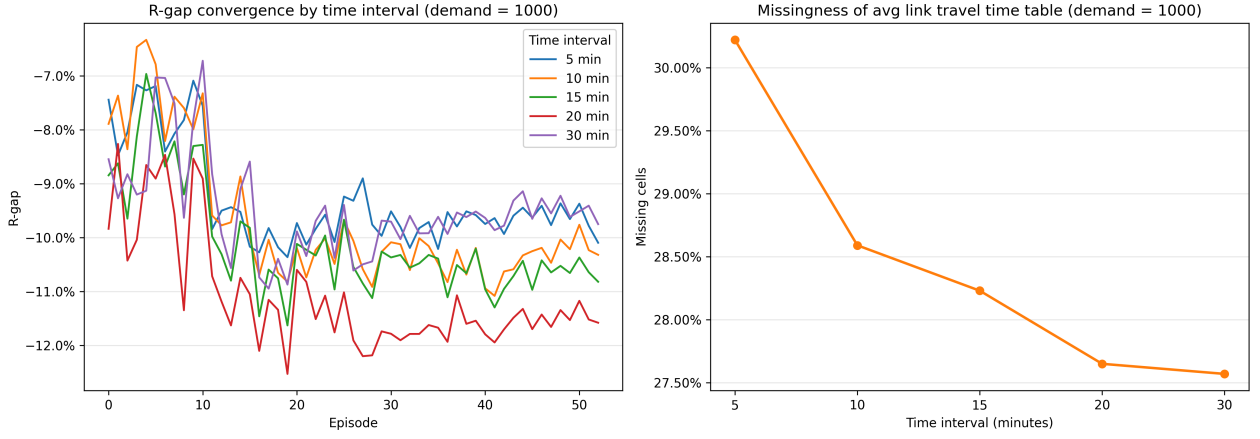


Figure 4: Sensitivity analysis fixed_time_min hyperparameter. **Low-demand: 1000**

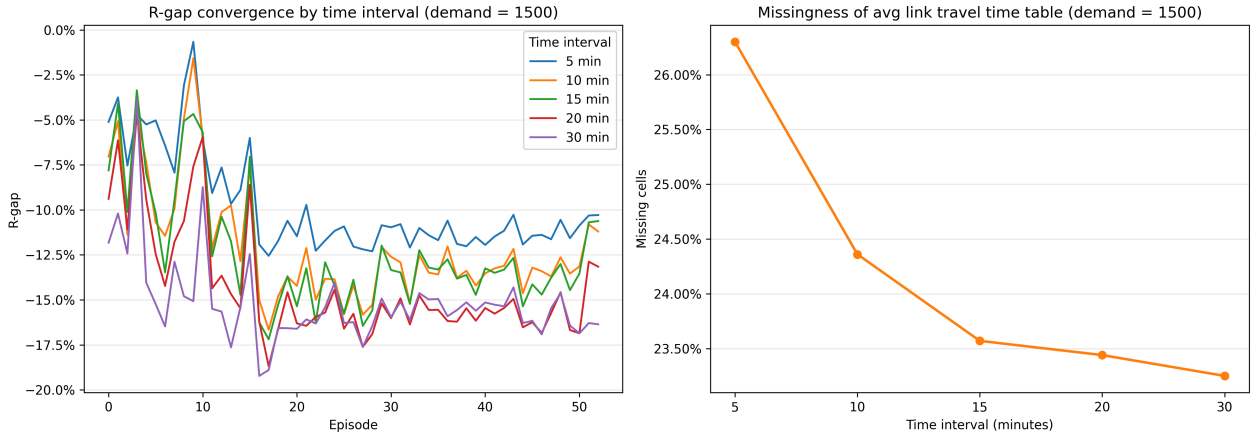


Figure 5: Sensitivity analysis fixed_time_min hyperparameter. **Low-medium demand: 1500**

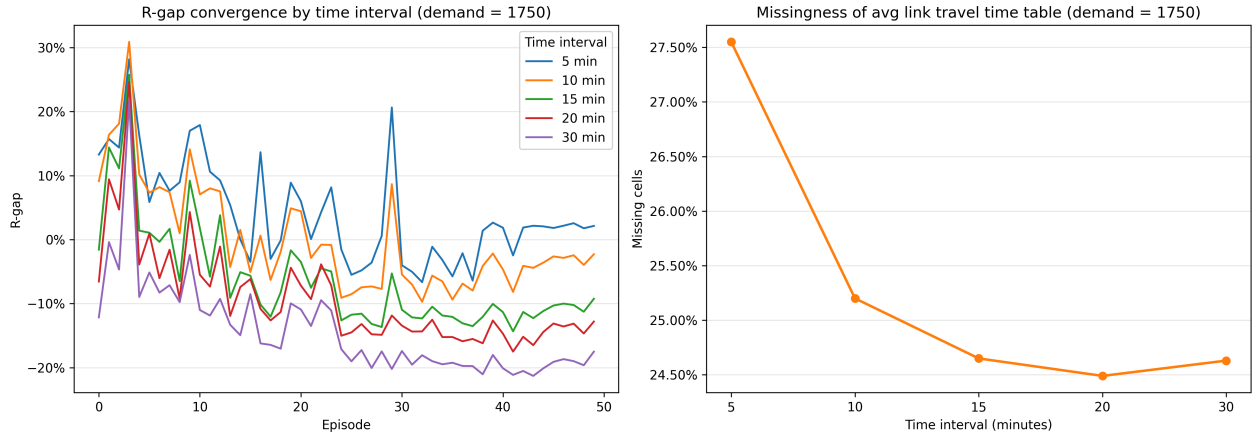


Figure 6: Sensitivity analysis fixed_time_min hyperparameter. **Medium-high demand: 1750**

3.3.2 Second group of plots

The second experiment explores the next time intervals: 30 seconds (0.5 min), 1, 5, 15, 30 and 60 minutes.

The results indicate that a 30-second interval provides the best overall performance. Across all demand scenarios, its R-gap curve remains consistently close to zero while rarely taking negative values.

The same general trend observed in the previous experiment is also present here: increasing the interval length progressively shifts the R-gap towards more negative values.

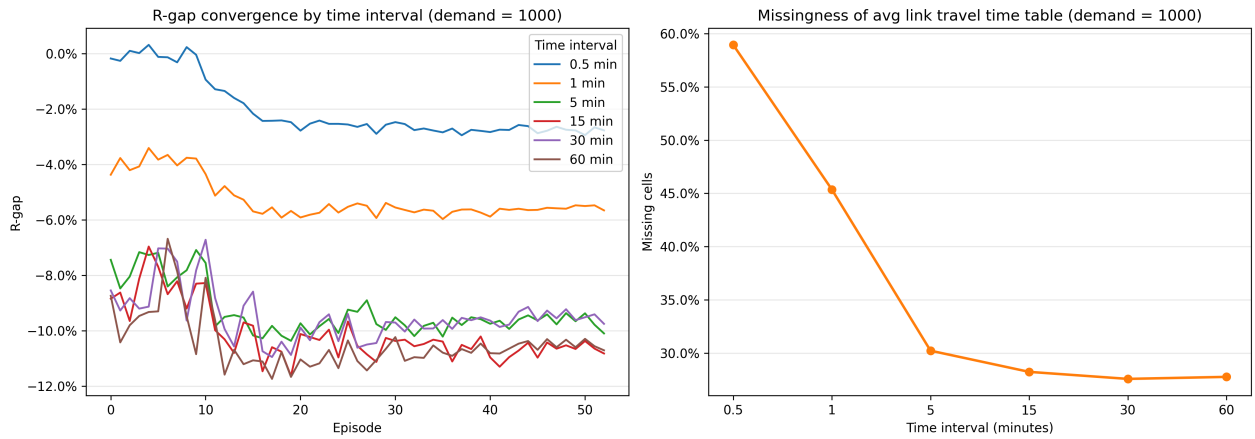


Figure 7: Sensitivity analysis fixed_time_min hyperparameter. **Low demand: 1000**

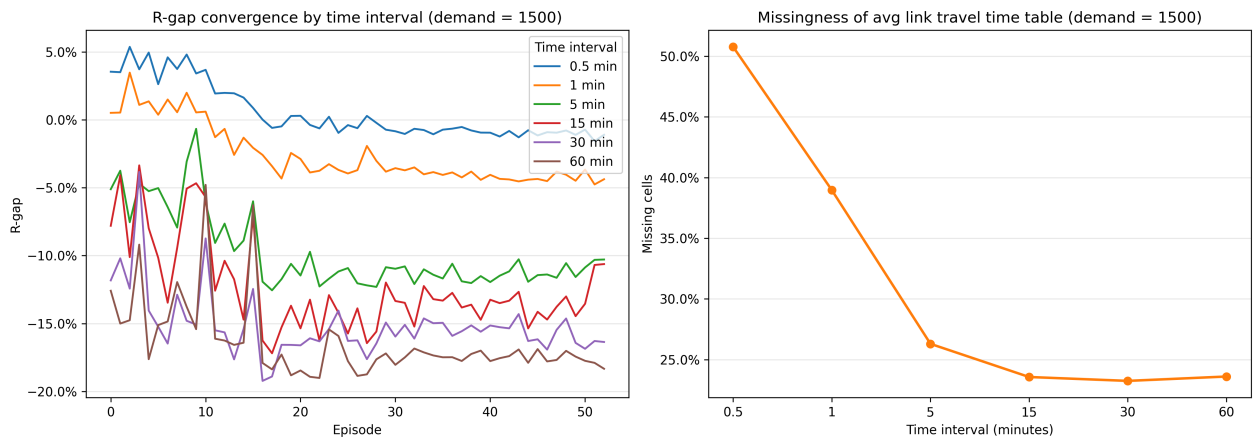


Figure 8: Sensitivity analysis fixed_time_min hyperparameter. **Low-medium demand: 1500**

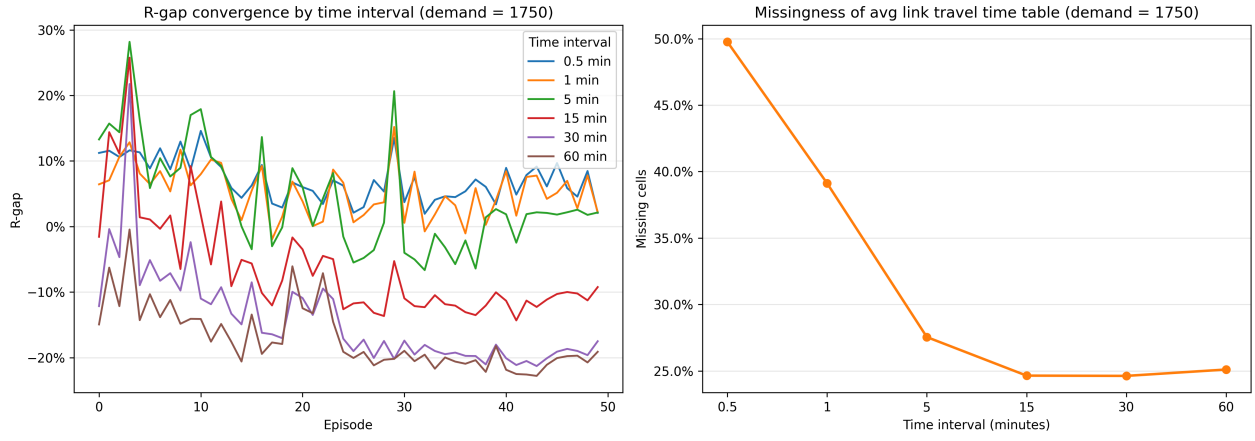


Figure 9: Sensitivity analysis fixed_time_min hyperparameter. **Medium-high demand: 1750**

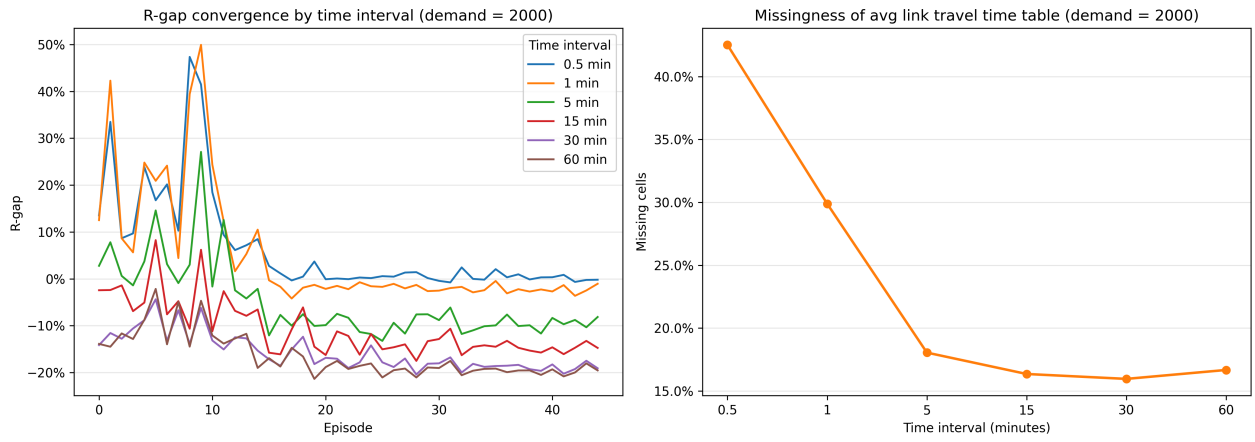


Figure 10: Sensitivity analysis fixed_time_min hyperparameter. **High demand: 2000**

3.3.3 Third group of plots

The final experiment further refines the search by considering intervals of 5, 10, 15 and 30 seconds.

The results indicate that both 15 and 30 seconds produce R-gap values very close to zero across all demand scenarios. Between these two candidates, a 15-second interval is preferred because it consistently produces positive R-gap values while remaining as close to zero as the 30-second interval.

Therefore, a fixed time interval of 15 seconds (0.25 minutes) is selected.

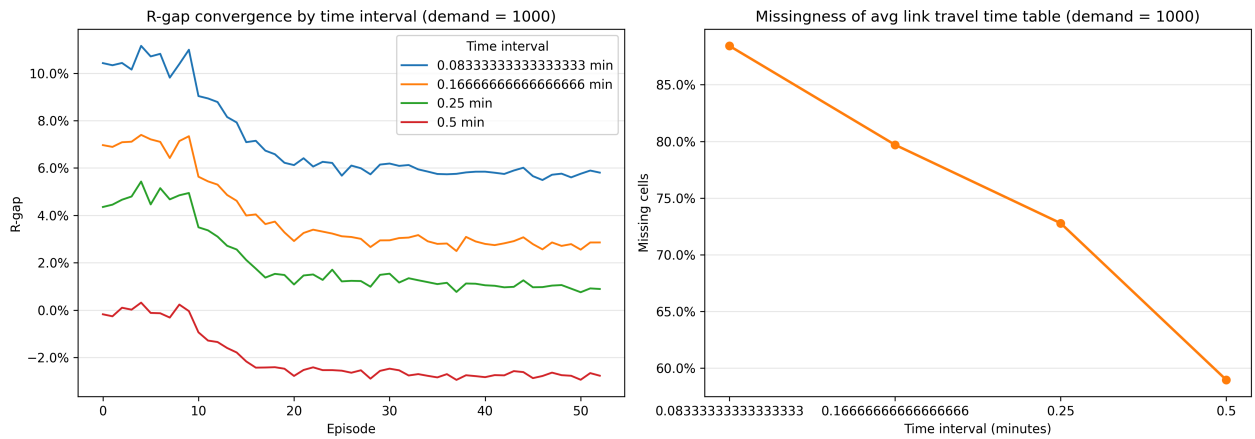


Figure 11: Sensitivity analysis fixed_time_min hyperparameter. **Low demand: 1000**

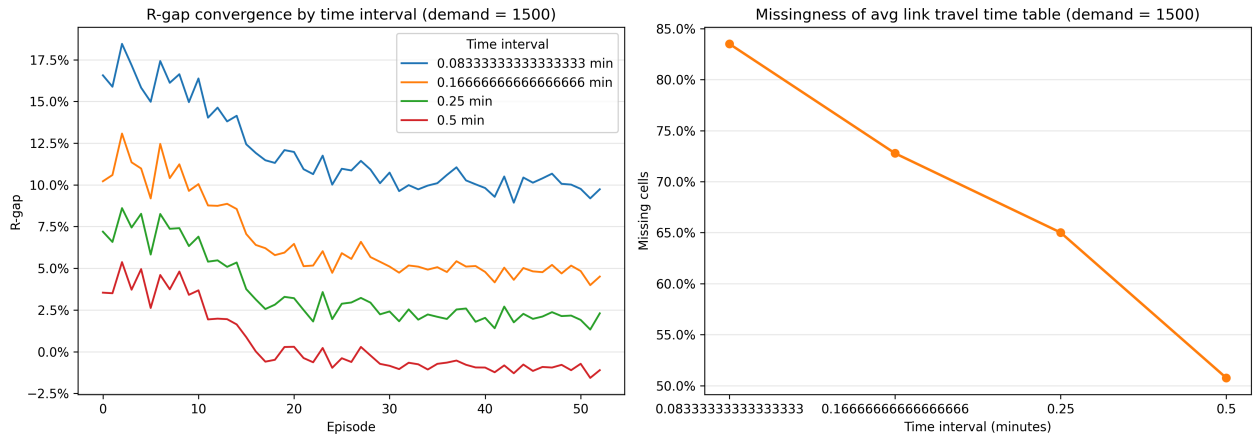


Figure 12: Sensitivity analysis fixed_time_min hyperparameter. **Low-medium demand: 1500**

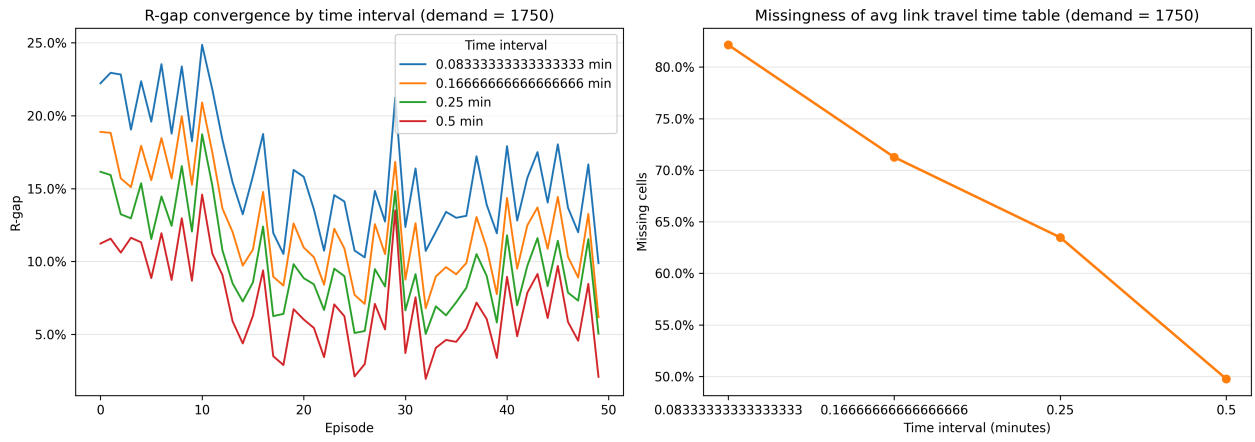


Figure 13: Sensitivity analysis fixed_time_min hyperparameter. **Medium-high demand: 1750**

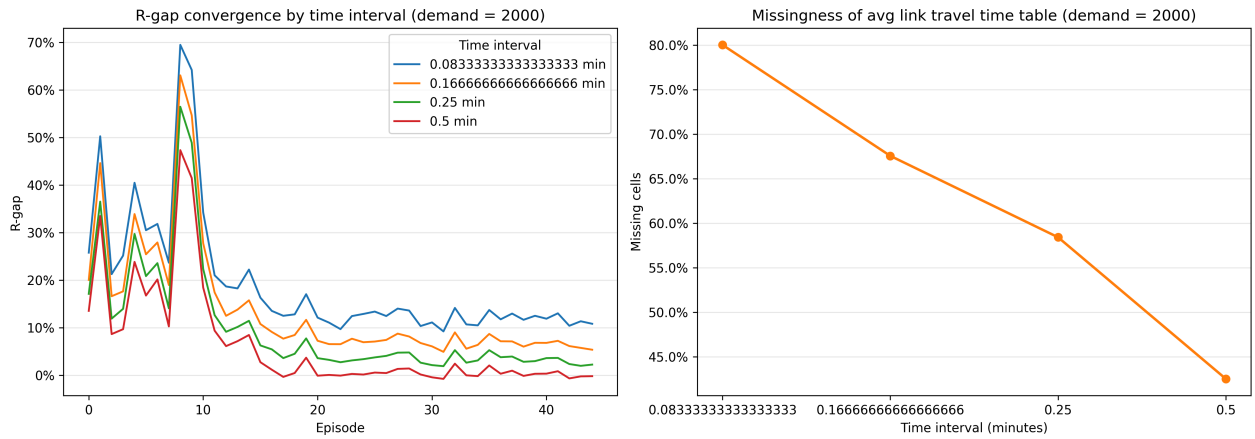


Figure 14: Sensitivity analysis fixed_time_min hyperparameter. **High demand: 2000**

Final decision 15 seconds